

A : FORMULARIUM of SIGNALS and SYSTEMS version 2

1. LAPLACE TRANSFORM

1.1 Definition and properties

<u>Definition</u> :	$f(t) \cdot u(t)$	o---o	$F(s) = \int\limits_{0^-}^{\infty} f(t) e^{-st} dt$
<u>translation in s</u> :	$f(t) e^{-at} \cdot u(t)$	o---o	$F(s + a)$
<u>translation in t</u> :	$f(t-a) \cdot u(t-a)$	o---o	$e^{-as} F(s)$
<u>derivative in t</u> :	$\frac{d}{dt} f(t) \cdot u(t)$	o---o	$s \cdot F(s) - f(0)$
	$\frac{d^2}{dt^2} f(t) \cdot u(t)$	o---o	$s^2 \cdot F(s) - s \cdot f(0) - f'(0)$
<u>derivative in s</u> :	$t \cdot f(t) \cdot u(t)$	o---o	$-\frac{d}{ds} F(s)$
	$t^n \cdot f(t) \cdot u(t)$	o---o	$(-1)^n \frac{d^n}{ds^n} F(s)$
<u>convolution</u> :	$f(t) * g(t) \cdot u(t)$	o---o	$F(s) \cdot G(s)$
<u>Scaling property</u> :	$f(a \cdot t)$	o---o	$\frac{1}{a} F\left(\frac{s}{a}\right)$
<u>Initial value theorem</u> :	$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$		
<u>Final value theorem</u>	$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s)$		
<u>Link with FTC</u> :	$LT(s = j\omega) = FTC(j\omega)$	if signal absolute integrable	

1.2 Useful Laplace pairs

$e^{-at} u(t)$	o---o	$\frac{1}{s+a}$	;	$t^n u(t)$	o---o	$\frac{n!}{s^{n+1}}$
$\cos(a.t) u(t)$	o---o	$\frac{s}{s^2 + a^2}$	;	$\sin(a.t) u(t)$	o---o	$\frac{a}{s^2 + a^2}$
$\delta(t)$	o---o	1	;	$u(t)$	o---o	$\frac{1}{s}$
$t \cos(a.t) u(t)$	o---o	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$	;	$t \sin(a.t) u(t)$	o---o	$\frac{2sa}{(s^2 + a^2)^2}$

2 FOURIER TRANSFORM (FTC)

2.1 Basic formulae

FTC :
$$X(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

Inverse FTC :
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) e^{j\omega t} d\omega$$

convolution theorem : in time : $f(t) * g(t) \longleftrightarrow F(\omega) \cdot G(\omega)$

in frequency : $f(t) \cdot g(t) \longleftrightarrow \frac{1}{2\pi} F(\omega) * G(\omega)$

translation : $x(t) \longleftrightarrow X(\omega)$ then $x(t-t_0) \longleftrightarrow X(\omega) e^{-j\omega t_0}$

symmetry : $X(-\omega) = X^*(\omega)$

time symmetry : if $x(t) \longleftrightarrow X(\omega)$ then $x(-t) \longleftrightarrow X(-\omega) = X^*(\omega)$

Link between Fourier series and Fourier transform : $X(k\omega_0) = T \cdot c_k$

2.2 Useful Fourier pairs

Block : $x(t) = A$ on $[-L/2, L/2]$ $\longleftrightarrow X(\omega) = AL \operatorname{sinc}(\omega L/2)$
 $= 0$ elsewhere

Sinc : $x(t) = A \operatorname{sinc}(\omega_0 t)$ $\longleftrightarrow X(\omega) = A\pi/\omega_0$ for $|\omega| < \omega_0$
 $= 0$ elsewhere

Impuls : $x(t) = \delta(t - t_0)$ $\longleftrightarrow X(\omega) = e^{-j\omega t_0}$

Complex expon. $x(t) = e^{j\omega_0 t}$ $\longleftrightarrow X(\omega) = 2\pi \delta(\omega - \omega_0)$

Cosine $x(t) = \cos(\omega_0 t)$ $\longleftrightarrow X(\omega) = \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$

Sine $x(t) = \sin(\omega_0 t)$ $\longleftrightarrow X(\omega) = j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

Delta train $x(t) = \sum_k \delta(t - kT)$ $\longleftrightarrow X(\omega) = \frac{2\pi}{T} \sum_k \delta(\omega - \frac{2\pi k}{T})$

3 FOURIER SERIES (FS)

3.1 Cartesian form

Signal
$$f(t) = \frac{a_0}{2} + \sum_{k=1}^{+\infty} \left[a_k \cdot \cos\left(\frac{2\pi \cdot k}{T} t\right) + b_k \cdot \sin\left(\frac{2\pi \cdot k}{T} t\right) \right]$$

Coefficients :
$$a_0 = \frac{2}{T} \int_0^T f(t) dt \quad a_k = \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi \cdot k}{T} t\right) dt \quad b_k = \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi \cdot k}{T} t\right) dt$$

3.2 Complex form

Signal
$$x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{j\omega_0 k t} \quad \text{with : } \omega_0 = 2\pi / T$$

Coefficients
$$c_k = \frac{1}{T} \int_0^T f(t) \cdot e^{-j\omega_0 k t} dt$$

symmetry :
$$c_{-k} = c_k^*$$

Spectrum
$$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} c_k \delta(\omega - k\omega_0)$$

3.3 Links between cartesian and complex form

$$c_k = \frac{a_k - j \cdot b_k}{2} \quad \text{and} \quad c_k^* = \frac{a_k + j \cdot b_k}{2}$$

$$|c_k| = \frac{1}{2} \sqrt{a_k^2 + b_k^2} \quad \text{and} \quad \phi_k = \text{Arctan2}(a_k, -b_k)$$

$$a_k = 2 \operatorname{Re} c_k \quad \text{and} \quad b_k = -2 \operatorname{Im} c_k$$